

# GAURAV KUMAR MAURYA

Computer Science Engineering | Software Development | AI/ML & Android

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LinkedIn | GitHub | LeetCode\_ | Portfolio

## PROFESSIONAL SUMMARY

Computer Science Engineering graduate with a strong foundation in Android application development using Kotlin and Android Studio. Hands-on experience developing Android applications and implementing on-device edge AI and machine learning models through academic and personal projects. Developed a Flutter-based real-time audio and video calling application using Firebase and ZEGOCLOUD, along with data-driven machine learning projects. Familiar with Python-based data science tools such as Pandas, NumPy, and Scikit-Learn, as well as foundational Generative AI concepts and architectures. Seeking an entry-level software/Android development role to apply and strengthen technical skills while contributing to robust and scalable solutions.

## TECHNICAL SKILLS

**Programming Languages:** Python, Kotlin, Java, SQL

**AI & Data Science:** Pandas, NumPy, Scikit-Learn, Machine Learning, Exploratory Data Analysis (EDA)

**Android Development:** Android Studio, CameraX API, Jetpack Compose, Room Database, SQLite

**Flutter Development:** Flutter, Dart, Firebase, Firestore, ZEGOCLOUD

**Generative AI:** LLM Fundamentals, RAG Fundamentals

**Version Control:** Git, GitHub

## PROJECTS

### ConnectCall: 1-to-1 Audio & Video Calling App

Sept 2026

Personal Project | Flutter · Dart · Firebase · Firestore · ZEGOCLOUD

- Real-Time Calling:** Developed a 1-to-1 audio and video calling application with outgoing/incoming calls, accept/reject, call termination, and real-time calling states using ZEGOCLOUD.
- Authentication & Backend:** Implemented email/password authentication and Firestore-based user management with searchable contacts and real-time online/offline presence.
- Call Controls & Error Handling:** Integrated microphone mute/unmute, camera on/off, camera switching, runtime permissions, and graceful handling of unavailable users, rejected calls, call failures, and permission issues.

### SafeStrideAI: AI-Based Real-Time Road Safety App

Jan 2026 – May 2026

Major College Project | Kotlin · Android Studio · TensorFlow Lite · OpenCV

- Edge AI & Model Optimization:** Integrated an on-device TensorFlow Lite object detection model for real-time pedestrian and vehicle detection at 20–30 FPS; applied INT8 quantization to reduce model size from 22 MB to 4.18 MB and improve inference performance.
- Computer Vision & Risk Detection:** Implemented Lucas-Kanade Sparse Optical Flow using OpenCV to estimate object motion and developed a Time-to-Collision (TTC) engine for real-time collision-risk assessment.
- Android Architecture & Alerts:** Designed a multi-threaded UI, inference, and database architecture with asynchronous Room/SQLite operations, and integrated real-time safety alerts with ARCore and Wear OS through the Wearable Data Layer API.

### Credit Card Default Prediction System

Aug 2025 – Oct 2025

Python · Pandas · NumPy · Scikit-Learn

- Supervised ML:** Built a machine learning pipeline predicting credit card defaults across 30,000 customer records.
- EDA:** Performed data wrangling, missing value handling, and feature engineering via Pandas and NumPy, identifying critical correlations between high credit utilization and late payment patterns.
- Model Selection:** Trained and cross-evaluated multiple Scikit-Learn classifiers, comparing accuracy vs. recall metrics to select the best-performing model.

## PROFESSIONAL EXPERIENCE

### Android App Development Intern: Internshala — Remote

Jul 2024 – Sep 2024

- Android Development:** Developed Android features and UI components using Kotlin and Android Studio based on assigned project requirements.
- Performance Optimization:** Used Android Studio Profiler to identify memory usage and UI performance issues and implemented optimizations to improve application responsiveness and stability.

## EDUCATION

### B.Tech – Computer Science & Engineering

2022 – 2026

Ch. Devi Lal State Institute of Engineering and Technology (CDLSIET), Haryana | Score: **CGPA: 8.96**

## CERTIFICATIONS

- Fundamentals of AI and Generative AI – Oracle (2025)
- Data Science & Machine Learning – YBI Foundation (2025)
- Python 101 for Data Science – IBM (2024)
- Android App Development Course – Internshala (2024)